

Type of organism

- 1) Eukaryotic
- 2) Pro karyotic Bacteria
- 3) Virus

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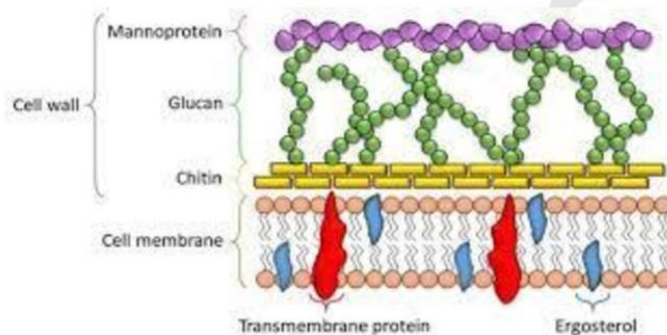
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Eukaryotic

Animal plant fungi, organism whose cell have a nucleus & other organelles enclosed within membrane

What is the Essential element of Fungai wall? **Chitin**

Makeup of cell membrane fungi? **Ergostrol**



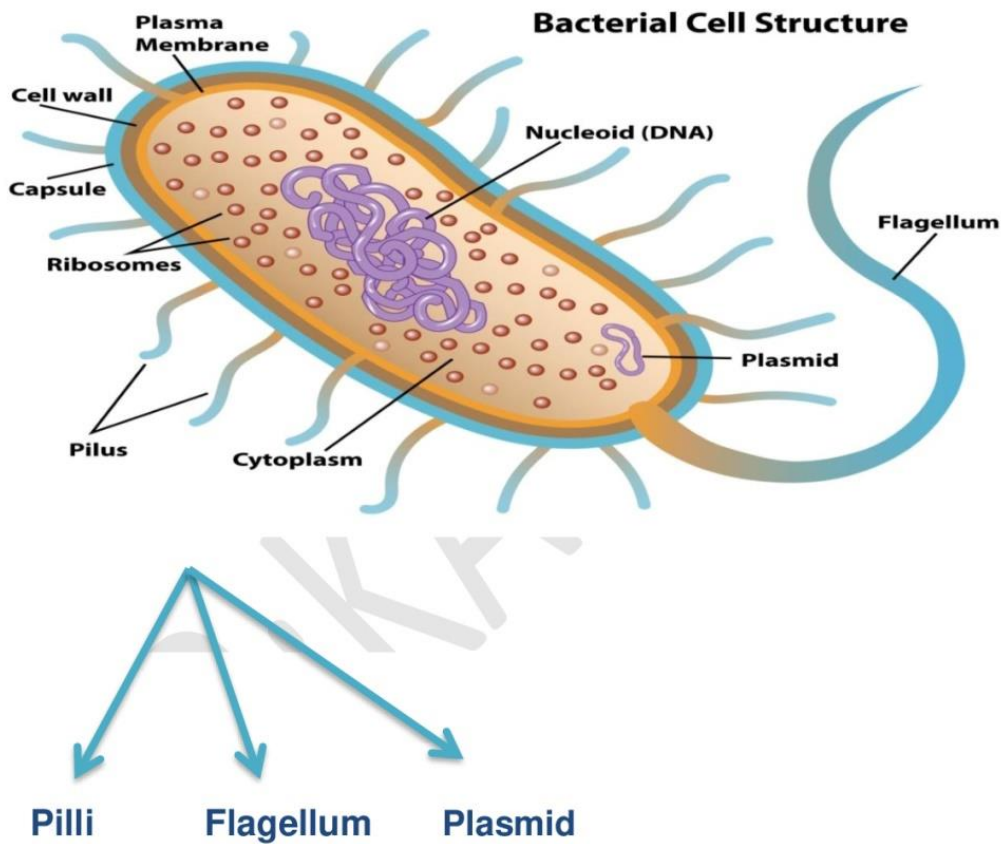
Pro karyotic

Bacteria, archea unicellular lack nucleus or any membrane bound organell ...divide by binary fission.

Virus:

Replicate only in living organism..Lack cell wall & membrane

Bacteria



Pilli:

Attachment & gene transfer during bacterial conjugation

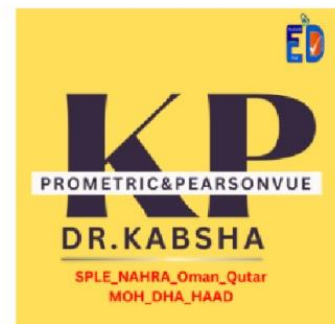
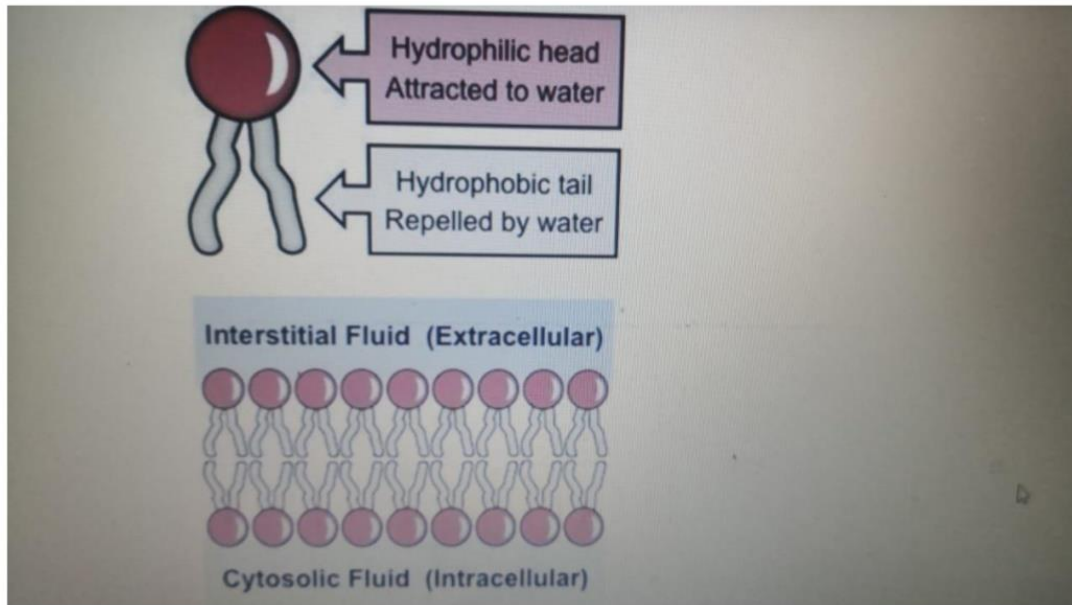
Flagellum

Mainly for movement

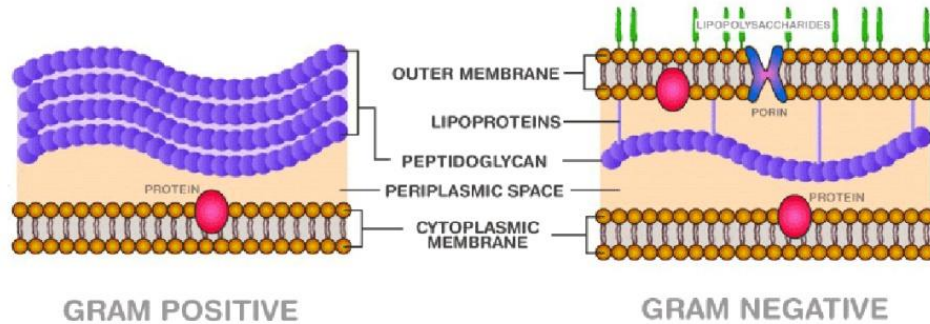
Plasmid

Small circular double strand DNA,, antibiotic resistance

phospholipid



GRAM-POSITIVE AND GRAM-NEGATIVE BACTERIA



Thickness	Thick	Thin
Peptidoglycan	Numerous layers of peptidoglycan (60%:90%) of the cell wall	2:3 layers of peptidoglycan (10%:20%) of the cell wall
Lipopolysaccharides	×	
Outer membrane	×	
Stain	crystal violet dye	Safranin" → Pink

Endotoxins heat stable „in gram **negative**

Exotoxin ??

Lack of outer membrane ?

Gram positive Bacteria

Cocci

Staphylococcus spp
Streptococcus spp
Enterococcus spp

Bacilli (Rods)

Non-Spore Forming (Aerobic)

Corynebacterium spp.
Erysipelothrix spp.
Nocardia spp.

Non-Spore Forming (Anaerobic)

Propionibacterium spp.
Actinomyces spp.
Listeria spp.
Lactobacillus spp.
Peptostreptococcus spp.
Bifidobacterium spp

Spore-forming (Aerobic)

Bacillus spp.

Spore forming (Anaerobic)

Clostridium spp

Atypical

Bacteria that do not color with gram staining

Chlamydia
Mycoplasma
Rickettsia

Mycobacteria

(Mycobacteria a genus of rod-shaped **acid-fast** bacteria having two significant pathogenic species)

-Mycobacterium

Lepae; leprosy

-Mycobacterium

Tuberculosis;
Tuberculosis:

Gram Negative Bacteria

Cocci

Neisseria gonorrhoeae
Neisseria meningitidis

Bacilli (Rods)

Escherichia coli
Vibrio cholerae
Klebsiella spp
Salmonella spp
Shigella spp
Proteus spp
Pseudomonas aeruginosa
Helicobacter pylori
Legionella pneumophila
Haemophilus influenza
Brucella spp
Bordetella pertussis

Pseudomonas aeruginosa:

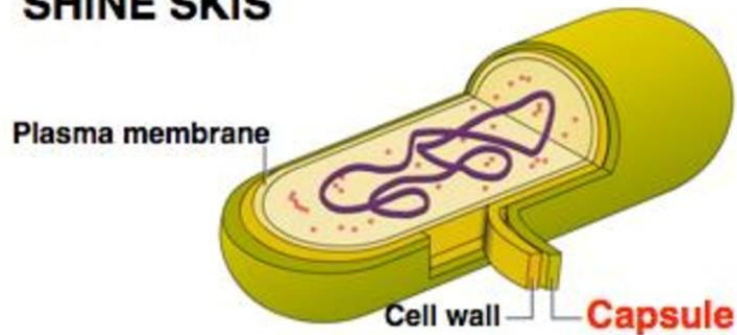
- a) Gram +ve cocci
- b) Gram -ve cocci
- c) Gram +ve bacilli
- d) Gram -ve rods

staphylococcus aureus → arranged in a grape like cluster



Encapsulated Bacteria

SHiNE SKiS



Streptococcus pneumoniae
Hemophilus influenzae type B
Neisseria meningitidis
Escherichia coli

Salmonella typhi
Klebsiella pneumoniae
Streptococcus (Group B)

Cryptococcus neoformans: encapsulated yeast

Environmental requirements for
microbial growth

<u>Obligate aerobes</u>	Growth occurs only in presence of O ₂	✓ Mycobacterium tuberculosis Pseudomonas
<u>Facultative anaerobes</u>	Growth occurs in presence or absence of O ₂ (better in O ₂)	✓ Most bacteria ✓ E. coli & Staph
<u>Obligate anaerobes</u>	Growth occurs only in absence of O ₂ (doesn't tolerate O ₂)	✓ Clostridium species Bacteroid , Trepomonas agilis
<u>Aerotolerant anaerobes</u>	Growth equally well in presence or absence O ₂	✓ Streptococcus pyogenes
<u>Microaerophile anaerobe</u>	Growth occurs only in presence of low O ₂ concentration.	✓ H. Pylori

Psychrophiles	0°C - 15 °C	→	Flavobacterium species
Mesophiles•	25°C - 40 °C	→	Pseudomonads aeruginosa & E. Coli & Staph
Moderate thermophiles	• 55°C - 65 °C	→	Bacillus stearothermophilus
Extreme thermophiles	• 80°C - 113 °C	→	Thermococcus celer

QUESTIONS:

1-Which of the following is a characteristic for facultative anaerobic?

- A) Only grow anaerobically
- B) Only grow in the presence of oxygen
- C) An anaerobe but can grow in the presence of oxygen
- D) Require simultaneous presence of aerobic and anaerobic environments for growth

2- Which of the following micro-organisms causes typhoid fever ?

- A) Shigella
- B) Flavivirus
- C) Salmonella
- D) Treponema pallidum

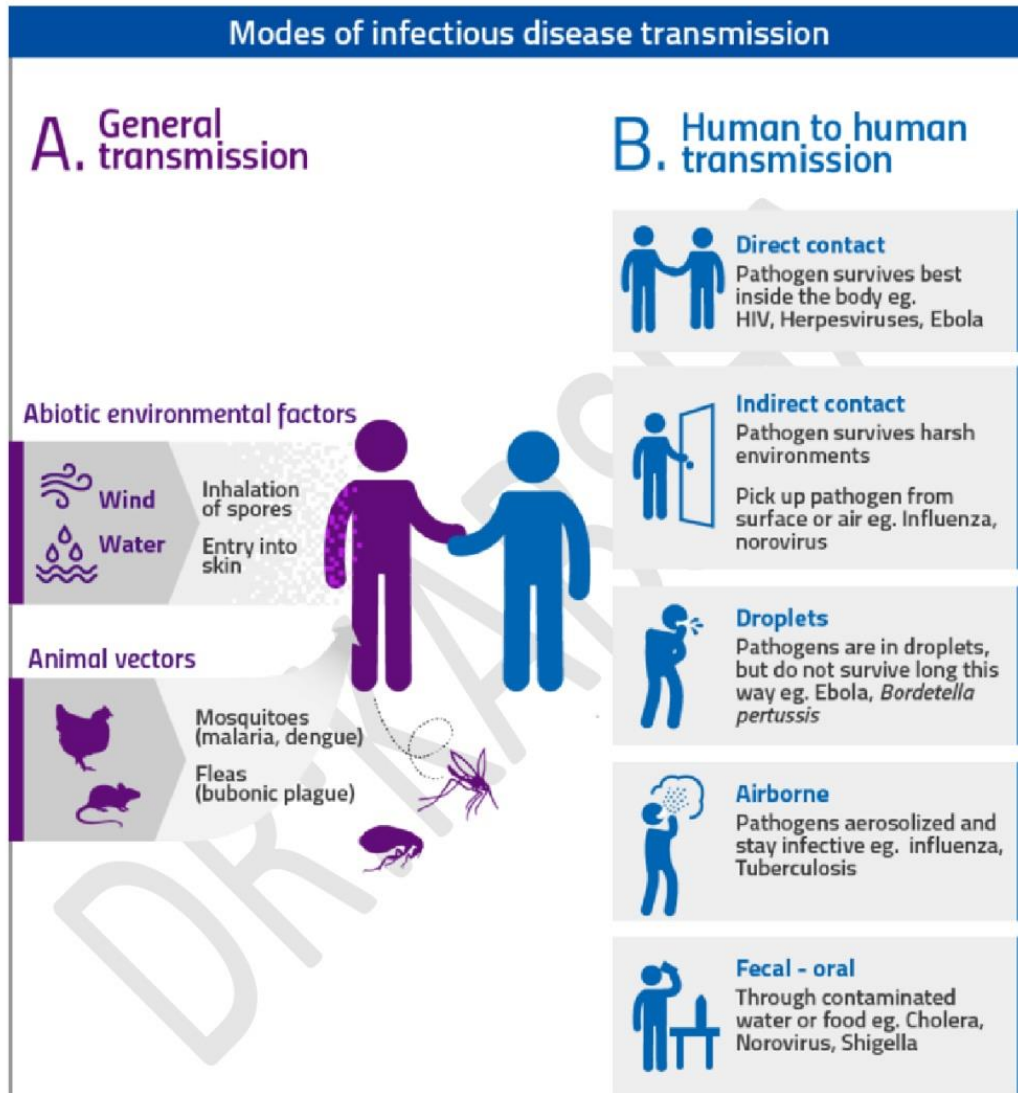
3- What is the main method transmission of Brucella species ?

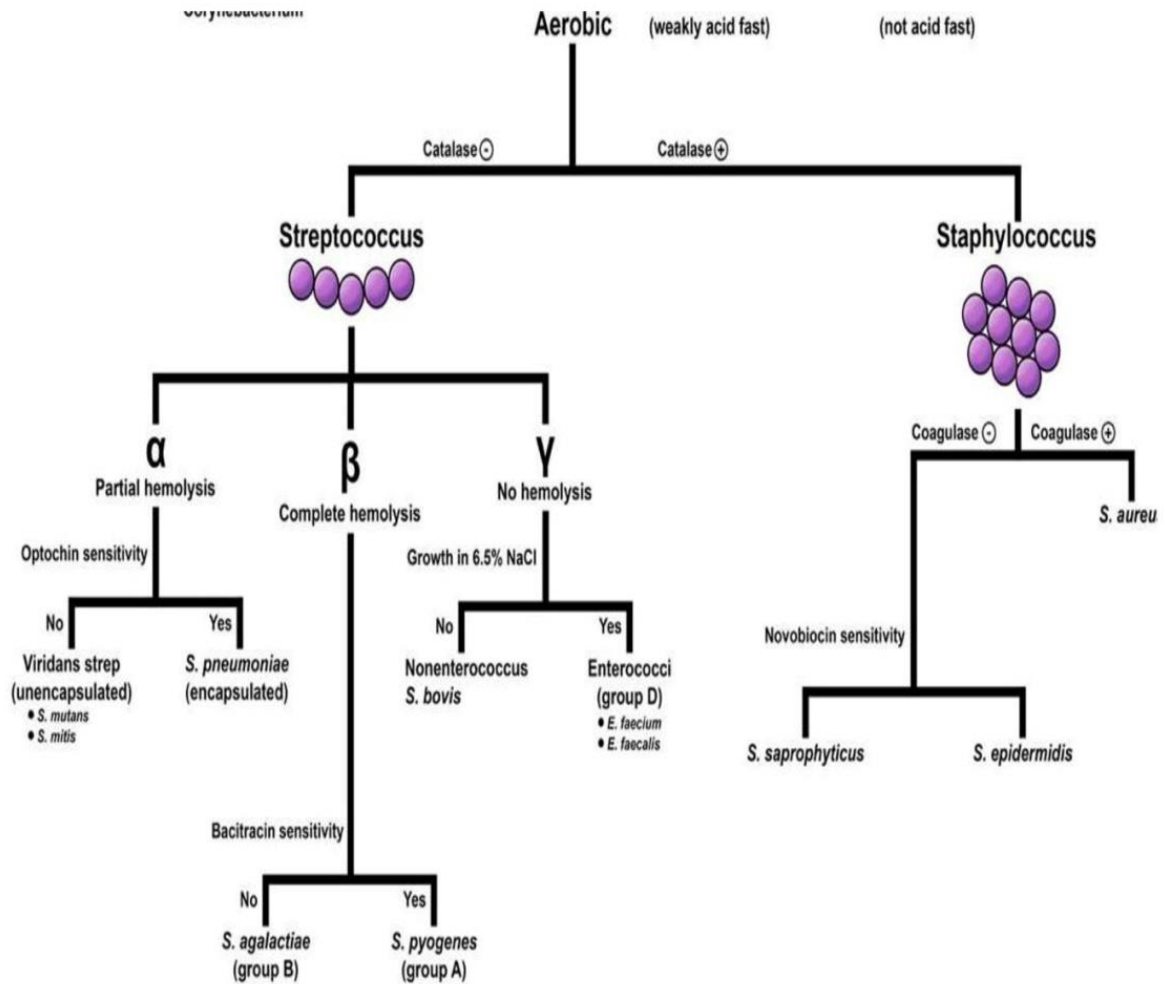
- A) A)Airborne
- B) Blood transfusion
- C) Contact with infected person
- D) ingestion of unpasteurized milk /milk products

- Brucella species?

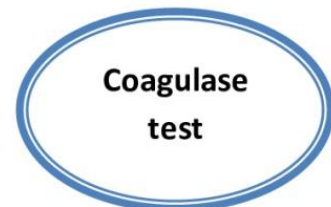
- Typhoid fever?

❖ Modes of transmission Ebola?? Hepatitis A?

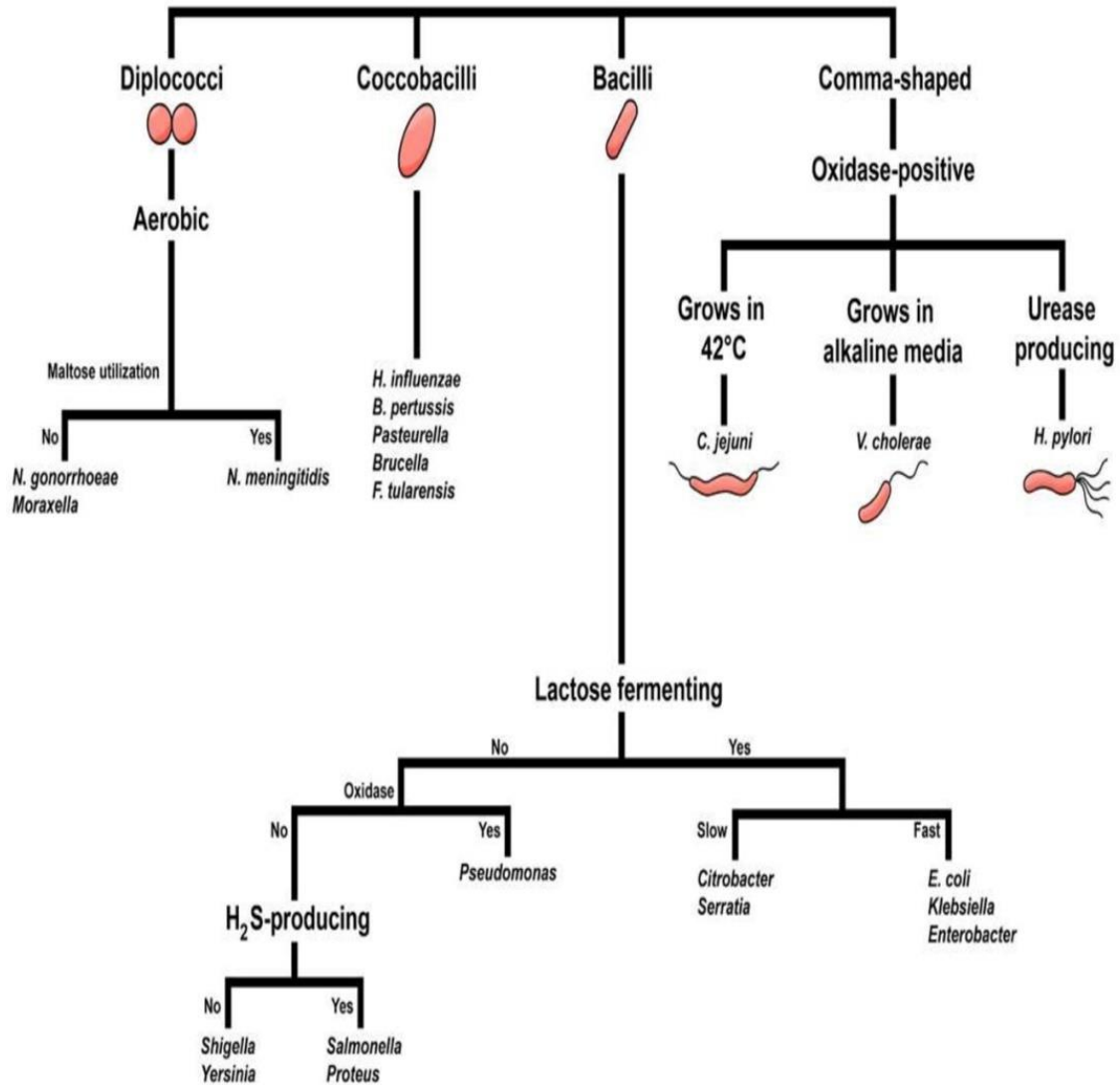




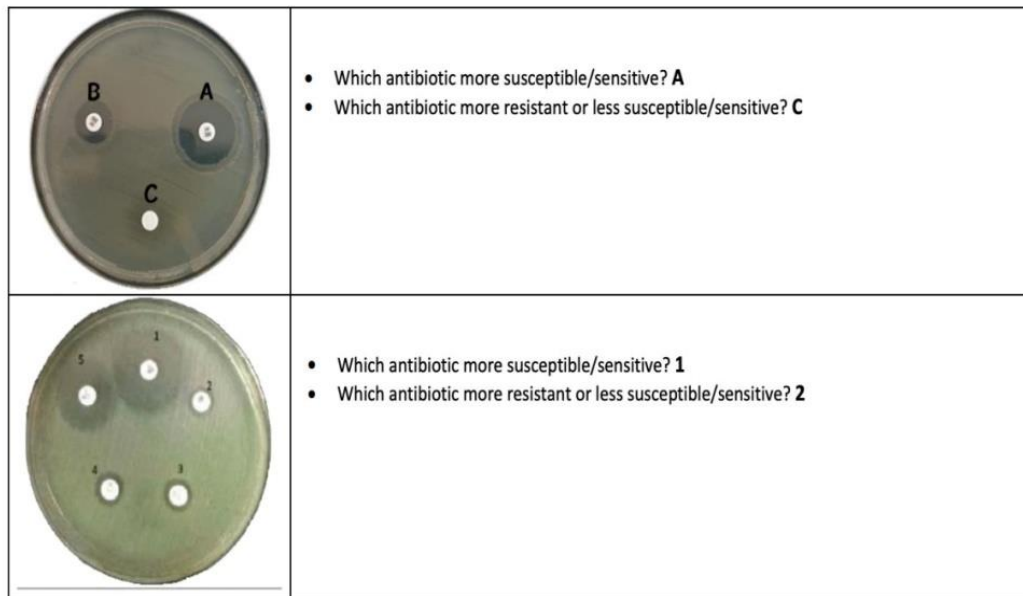
Differentiate between *s.epidermidis* & *s.aureus* by



Gram-Negative Bacteria



✚ **Diffusion Method – Kirby-Bauer Method:** The bigger the circle the greater the sensitivity

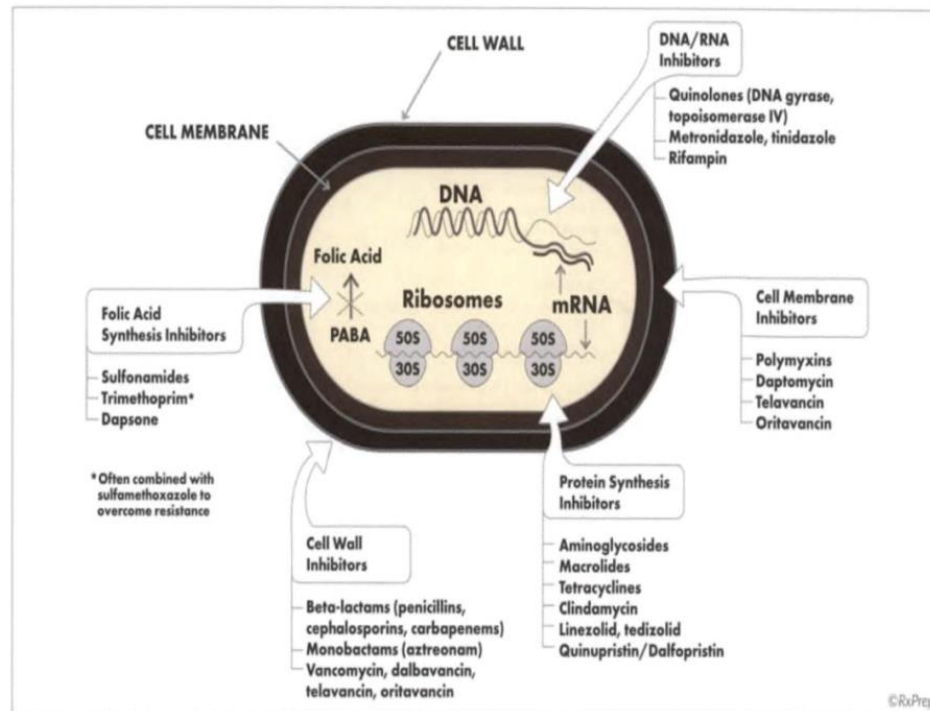


Antibiotic Stewardship: coordinated program that promotes the appropriate use of antimicrobials (including antibiotics), improves patient outcomes, reduces microbial resistance, and decreases the spread of infections caused by multidrug-resistant organisms.

antibiogram aid in selecting empiric treatment

combines culture data from patients at a single institution into one chart (such as all Gram-positive organisms cultured at a hospital) to show susceptibility patterns over a specific time period (generally 1year).





∞ **Bacteriostatic M.S COLT**

∞ **Macrolide**

∞ **Sulfonamide**

∞ **Chrolamphenicol**

∞ **Oxazoldine**

∞ **Lincosamide (clindamycin)**

∞ **TTC**

Time dependent:

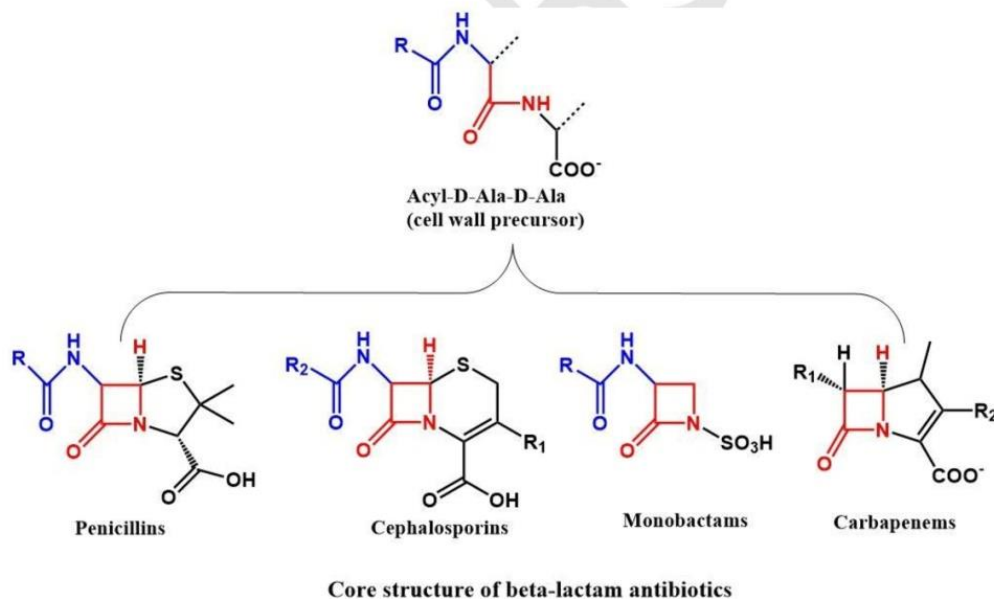
- Penicillin - Cephalosporins- Carbapenem - Vancomycin - Macrolide - Clindamycin
- dosed more frequently to maximize the time above the MIC ((MIC = Minimum inhibitory concentration))

Concentration dependent:

- Quinolone - Aminoglycoside - Azithromycin - Ketolides - Daptomycin
- dosed less frequently and in higher doses to maximize the concentration above the MIC.

ΩBetalactam antibiotics

Ω Penicillin



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They inhibit bacterial cell wall synthesis, time-dependent

Probenecid can increase levels of beta-lactams by interfering with renal excretion.

SIDE EFFECTS

- 1) Anaphylactic shock ,cross allergy
 - 2) Super infection (candidiasis & pseudomembranous colitis)
- ✚ Pseudomembranous colitis (clostridium difficile) treated by **oral** Vancomycin or metronidazole

Penicillin G (parenteral)

Uses ↓

- 1) Drug of choice for syphilis (*tropenema palledum*)
- 2) Meningitis (pass if inflamed)

Penicillin v oral short acting

Long acting

Procaine

Benzathine

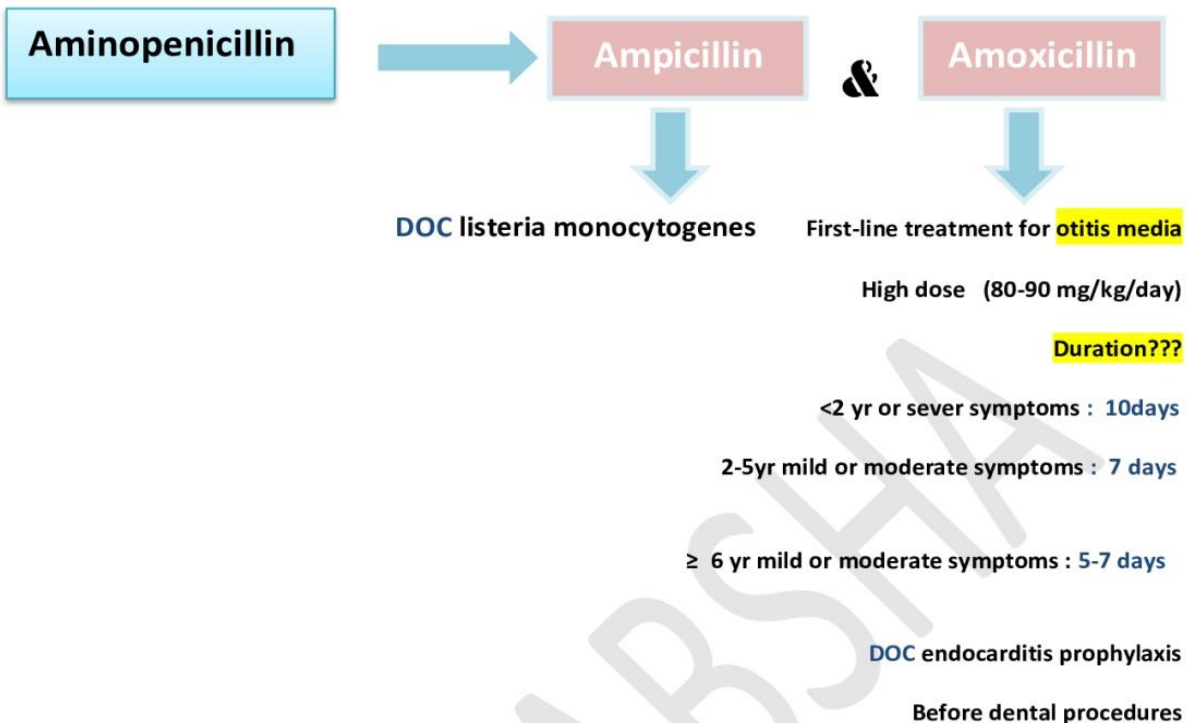
Penicillin in



rheumatic fever prophylaxis

Penicillinase resistant (ant staphylococcal)

- ➡ Methicillin, Nafcillin parenteral, **oral** oxacillin and dicloxacillin (No renal dose adjustment needed)
- ➡ Methicillin. Interstitial nephritis
- ➡ Nafcillin. Staphylococcal endocarditis



(Amino penicillin +B lactamase inhibitor) broad spectrum

- **Ampicillin** /sulbactam shigella ttt
 - **Amoxicillin** / clavulanic
- Otitis media

Against Pseudomonas

- **pipracillin** /tazobactam
- **ticarcillin** carbenicillin?

cephalosporin

1th Generation:	2th Generation:	3th Generation:	4th Generation	5th Generation
Cefazolin Cephalexin Cephradine Cefadroxil	Cefuroxime BBB Cefotetan } Cefoxitine } anerob cefmetazole }	ceftazidim,cefoperazon Ceftriaxone Ceftaxime Cefdinir	Cefepime Cefpirome	Ceftaroline Ceftobiprole Ceftolozane
Cefazolin parental in (CLABSI) Ttt				
BBBxx	Cefuroxime cross BBB	Cross BBB	Cross BBB	Cross BBB
	Colorectal	Antipseudomonous : ceftazidim, Cefoperazone (testicular atrophy) Ceftriaxone C.I in neonate 0-28 day (bilirubin) Ceftriaxone., Cefoperazone Safe in renal Ceftriaxone 2g BID in meningitis	Antipseudo monous :	Antipseudomonous : Ceftaroline The only beta- lactam that is active against MRSA

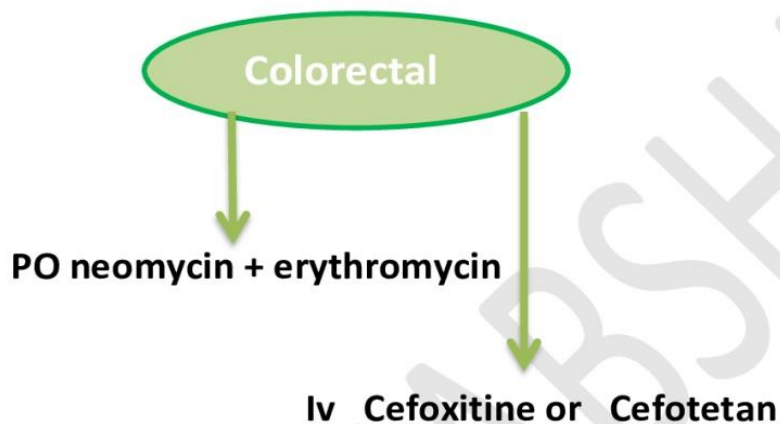
Structure oral cephalo????????????????????

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CLABSI (central line associated blood stream infection)

➔ Parentral anti staph nafcillin +cefazolin or vancomycin in case of MRSA
Fungi capsufangin , micafungin . anidulafungin safe in hepatic



- + **Meningitis** Ceftriaxone 2g BID
 - + **Otitis media** High dose (80-90 mg/kg/day) , amoxicillin First-line treatment
 - + **faecalis and E. faecium** are naturally (intrinsically) resistant to cephalosporins
- Ampicillin** is the drug of choice for monotherapy of susceptible *E. faecalis* infection

Meningitis

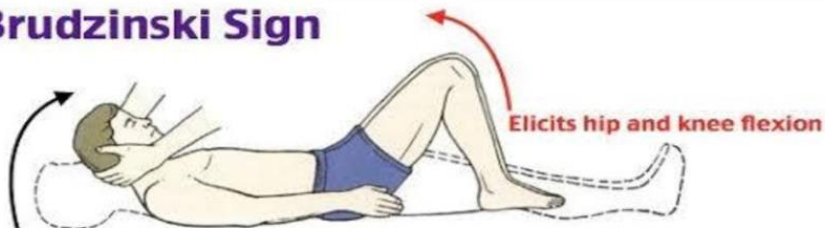
< 1 month	→	penicillin + cefotaxime (or AMG)
1month -50 year's	→	ceftriaxone +vancomycin
Above 50 years	→	ceftriaxone +vancomycin + ampicillin

Kernig Sign



- 1 Knee is flexed to 90 degrees
- 2 Hip is flexed to 90 degrees
- 3 Extension of the knee is painful or limited in extension

Brudzinski Sign



- 1 Passive flexion of neck

❖ cellulitis

What is the MDR Organism that could be the cause of purulent cellulitis?

MRSA

If systemic signs or multiple sites (moderate infection), perform culture fluid and use oral antibiotics that cover **MRSA**:

- **SMX/TMP** OS 1-2 tablets PO BID
- **Doxycycline** 100 mg PO BID
- **Minocycline** 200 mg PO x 1, then 100 mg PO BID
- **Clindamycin** 300 mg PO QID

Endocarditis

- Endocarditis by *staph.aerus*

Ampicillin + ceftriaxone

Penicillin G + gentamycin

Vancomycin + gentamycin

osteomyelitis

Pathogen : *staphylococcal*

Ttt : cefazolin ,naphcillin in resistance vancomycin or clindamycin

Pt with sickle cell anemia : ceftriaxone or cefotaxime

Duration:

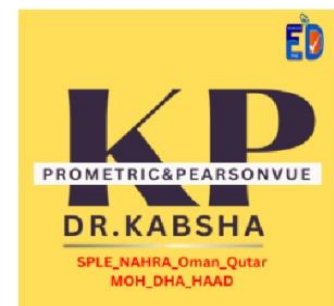
Acute osteomyelitis : 4-6 week

Chronic osteomyelitis : 6-8 week parenteral 3-12 month oral

If *p.aeruginosa*

mild : 500mg PO /12 hr

sever : Ciprofloxacin 750 mg PO BID or levofloxacin 750 mg PO q24h



Osteomyelitis

Definition	Infection of the bone with subsequent bone destruction
Signs and symptoms	a. Fever and chills b. Localized pain, tenderness, and swelling c. Neurologic symptoms if spinal cord compression
Laboratory tests	a. Elevated WBC b. Elevated erythrocyte sedimentation rate c. Elevated C-reactive protein
Diagnostic tests	a. Radiographic tests: Positive results lag behind infectious process. b. Computed tomography and magnetic resonance imaging scans c. Radionuclide imaging: Positive as soon as 24–48 hours after infectious process begins.

Empiric Therapy

1- Pediatric (neonate therapy should be tailored to the patient)	a. Cefazolin b. Nafcillin, oxacillin c. Clindamycin (use if prevalence of MRSA in community is 10% or more) d. Vancomycin (use if prevalence of MRSA and clindamycin-resistant <i>S. aureus</i> in community is 10% or more)
2- Adult	a. Nafcillin, oxacillin, cefazolin, ceftriaxone, clindamycin, or vancomycin (alternatives linezolid or daptomycin) b. Choose additional antibiotics according to patient-specific characteristics
3- Patients with sickle cell anemia	Ceftriaxone/cefotaxime or ciprofloxacin/levofloxacin (no studies assessing best empiric therapy)

4- Prosthetic joint infections

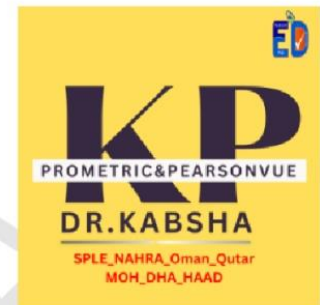
a. Debridement and retention of prosthesis or one-stage exchange of prosthesis	i. Staphylococcal: Pathogen-specific intravenous therapy plus rifampin 300–450 mg twice daily for 2–6 weeks, followed by rifampin plus ciprofloxacin or levofloxacin for 3 months (hip, elbow, shoulder, ankle prosthesis) or 6 months (knee prosthesis) ii. Non-staphylococcal: Pathogen-specific intravenous (or highly bioavailable oral) therapy for 4–6 weeks, followed by indefinite oral suppression therapy
b. Resection of prosthesis with or without planned reimplantation or amputation	a. Pathogen-specific intravenous (or highly bioavailable oral) therapy for 4–6 weeks b. Only 24–48 hours of antibiotic therapy after amputation if all infected tissue is removed

Therapy Length

Acute osteomyelitis: 4–6 week
Chronic osteomyelitis: 6–8 weeks of parenteral therapy and 3–12 months of oral therapy

Criteria for Effective Oral Therapy for Osteomyelitis

- | | | |
|---|------------------------------------|------------------------------|
| 1- High bioavailability antibiotic is available | 2- Adherence | 3- Resolving clinical course |
| 4- Identified organism that is highly susceptible to the oral antibiotic used | | |
| 5- C-reactive protein less than 2.0 mg/dL | 6- Adequate surgical debridement , | |



• **Monobactam**

★ **Aztreonam**

similar to beta-lactams; it inhibits bacterial cell wall synthesis by binding to penicillin-binding proteins (PBPs), which prevents the final step of peptidoglycan synthesis in bacterial cell walls. Aztreonam covers many Gram-negative organisms, including Pseudomonas. It has no Gram-positive or anaerobic activity. x x x x x

No cross sensitivity

• **Carbapenems**

★ **Meropenem - Ertapenem - imipenem /cilastatin**

Very broad-spectrum antibiotics except ertapenem not active against

Pseudomonas, acinetobacter , enterococci

Require dose adjustment (seizure) ...ertapenem less risk

Can cross csf if inflamed

- Imipenem compounded with cilastatin to protect it from metabolism by renal dehydropeptidase

Aztreonam

-ve ,pseudomonas not +ve aneorobes

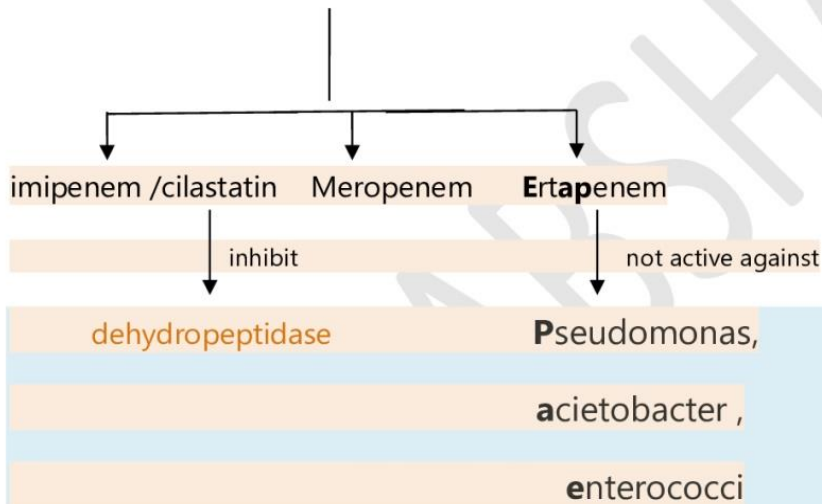
BBB

No cross hypersensitivity



Carbapenems seizure risk,, renal dose adjustment

- **Broad spectrum**(active against ESBL-producing organisms) but **not** Atypicals, VRE, MRSA, C. difficile,)



Non B lactam acting on cell wall

- **Vancomycin**

★ MOA: Inhibits bacterial cell wall synthesis by binding to the D-alanyl - D-alanine cell wall precursor and blocking peptidoglycan polymerization

Only against gram +ve „„MRSA & MRSE

Oral

Low bioavailability act locally

Treat pseudomembranous colitis caused by clostridium difficile

Iv infusion

Treat MRSA & MRSE

S/E red man syndrome 2ml /min

Increase infusion time = decrease infusion rate

Ototoxicity „ nephrotoxicity

Trough Level is: 10-15

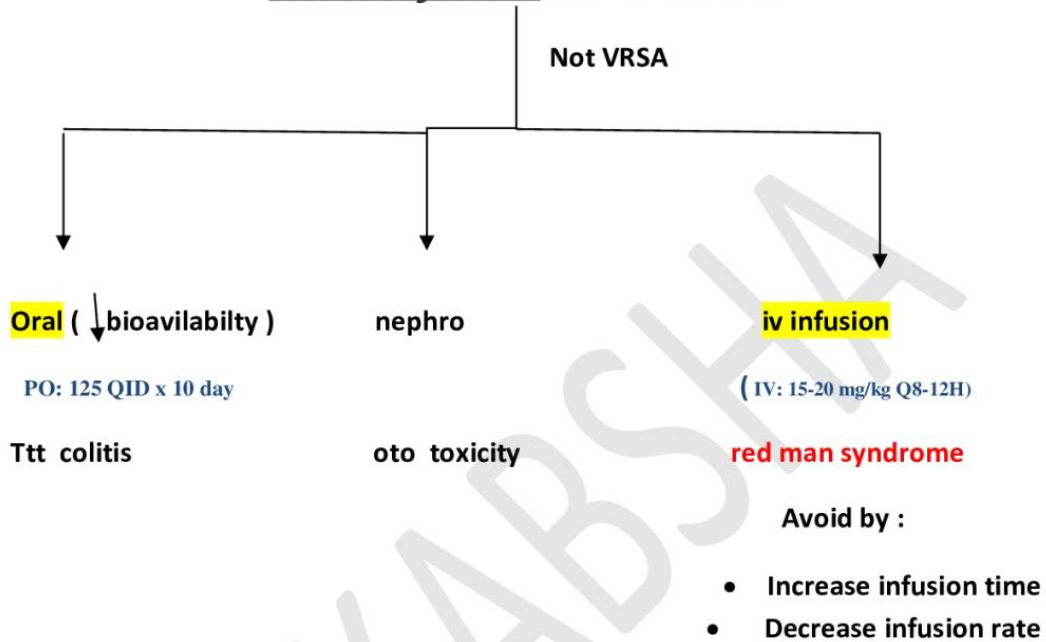
In severe infection 15-20mg /l

Vancomycin trough levels should ideally be drawn immediately before administration of the 4th dose

- How to make the highest difference between peak and trough ?

Non B lactam acting on cell wall

Vancomycin # MRSA ,MRSE



Cell membrane disruption

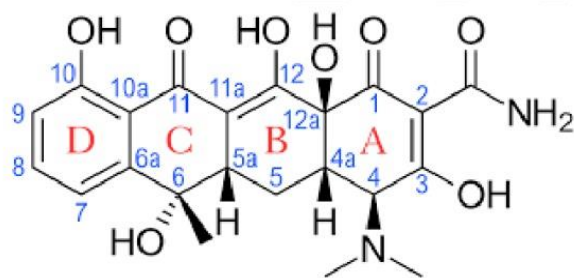
→ Daptomycin

VRSA ... gram +ve

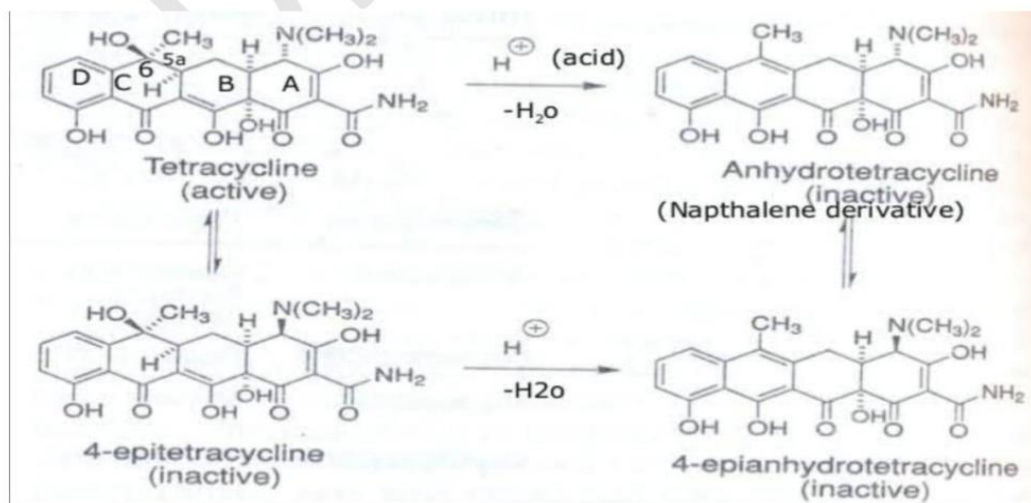
- Not used with statin
- drug inactivated in the lung by surfactant

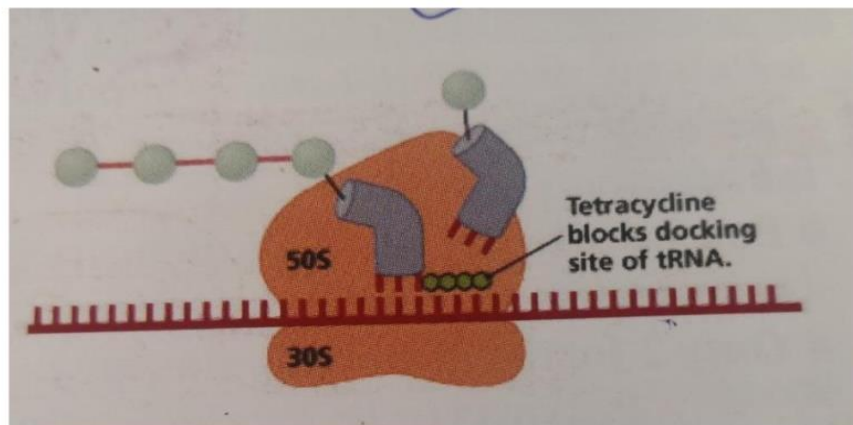
→ Polymyxin B ,,-ve neuro,nephrotoxic

TTC



In acid?





Doxycycline – Minocycline CSF

→ Doxycycline safe in renal & hepatic pt (no need for renal adjustment)

MOA : Binding to 30S subunit block tRNA binding site

DOC in

- Atypical
- Rickettsia , mycoplasma , Chlamydia
- Lyme , cholera

S/E

- photosensitivity
- Fanconi syndrome

C.I in pregnancy, breast feeding , children less than 8 year ..teeth discoloration

- Tablets and capsules should be taken with 8 oz. of water with doxycycline, upright for at least 30 minutes after dose

(Glycylcycline) Tigecycline

Not active against ppp..pseudomonas , proteus , providencia

Safe in renal but need adjustment in hepatic

Tigecycline & warfarin ↓ CL of warfarin

• **Aminoglycosides** **Gentamicin - Tobramycin - Amikacin - Streptomycin**

MOA : inhibit protein synthesis by irreversibly binding to **30S**

Bactericidal ...conc dependent

gram -ve (pseudomonas)

Combination AMG + B lactam in MRSA

Ttt ... plague caused by yersinia pestis

,,, tularemia (pneumonia with rabbit hunting)

S/E nephro, otot toxicity likes???

Calculation:

1- Administer 7-mg/kg gentamicin with initial dosage interval:

Estimated Crcl	Initial Dosage Interval
≥60 ml/min	Q24h
40_59 ml/min	Q36h
20_39 ml/min	Q48h
<20 ml/min	Monitor serial concentrations and administer next dose when < 1mg/ml

2- Obtain timed serum concentration, 6-14 hours after dose ideally first dose)

3- Alter dosage interval to that indicated by the Nomogram zone

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Hartford Nomogram Example

- MJ is a 50 year old female who has just been diagnosed with pneumonia secondary to pseudomonas aeruginosa. The provider would like to add tobramycin to MJ's current regimen.
- Actual body weight : 70 kg
- Height: 67
- Ideal body weight : 61.6kg
- SCr 0.9 mg/dl
- Pseudomonas MIC for tobramycin = 1mcg/ ml

Step 1

- Determine which body weight to use
- Actual body weight is not 1.2x's greater than ideal body weight, so use actual weight of 70 kg

Step 2

- Calculate dose
- Dose tobramycin = (7 mg/kg) (70kg) = 490 mg

Step 3

- Draw tobramycin level 6_14 h after dose given
- Level drawn after 10 hours and comes back at 6 mcg/ml

Monitoring for extended interval dosing

❖ Hartford Nomogram

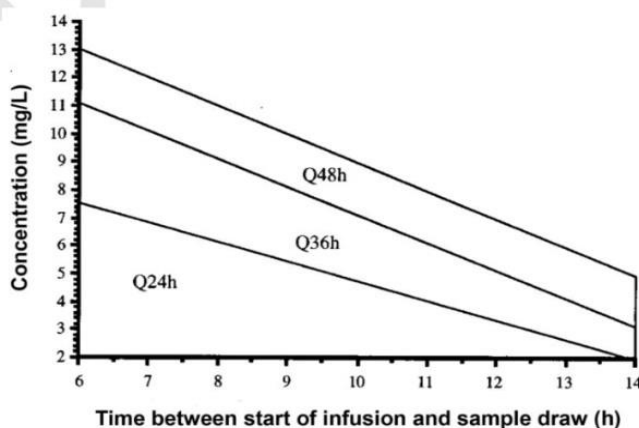
- Draw serum level 6_14 h after the first dose
- Used to determine dosing interval

❖ Check repeat level in 2_3 days or sooner if:

- Changing renal function
- Changing fluid status

❖ For patients on amikacin:

- Need to divide level by correction factor of 2 when using the Hartford Nomogram



• Chloramphenicol :

50S

- Gray`baby syndrome neonates due to low capacity of glucuronidation ??
- Bone marrow suppression
- Hemolytic anenmia in G6PD deficiency (favism)

Marcolide good alternative for b lactam allergy

MOA: 50S,,ihibitprotein synthesis

S/E : QT prolongation ototoxicity,,hepatotoxicity

Azithromycin for COPD exacerbations, distribute in most tissue except CSF

drug of choice for dysentery (travelers' diarrhea with bloody stools).

atypical (Chlamydia)

Azithromycin, erythromycin safe in renal

Erythromycin iv has prokinetic effect

Spiramycin in toxoplasmosis in pregnancy

With statin ??

❖ ttt of pertussis:

Erythromycin 40-50 mg/kg/day for 14 d or

Clarithromycin 15mg/kg for 7d or

Azithromycin 10mg/kg for 5 d

• Oxazolidinones linezolid, tedizolid

coverage similar to vancomycin, but also cover VRE (E. faecium and E. faecalis).

MRSA, VRSA, VRE

➔ Advantage over daptomycin & vancomycin

- Safe in renal & hepatic
- Category C in pregnancy & children use

S/E serotonin syndrome with non selective MAOI

Optic neuropathy

> 14 days, thrombocytopenia

Monitor CBC

Lincosamide (clindamycin)

Safe in renal # anaerobic

Accumulate in bone teeth urine but not CSF

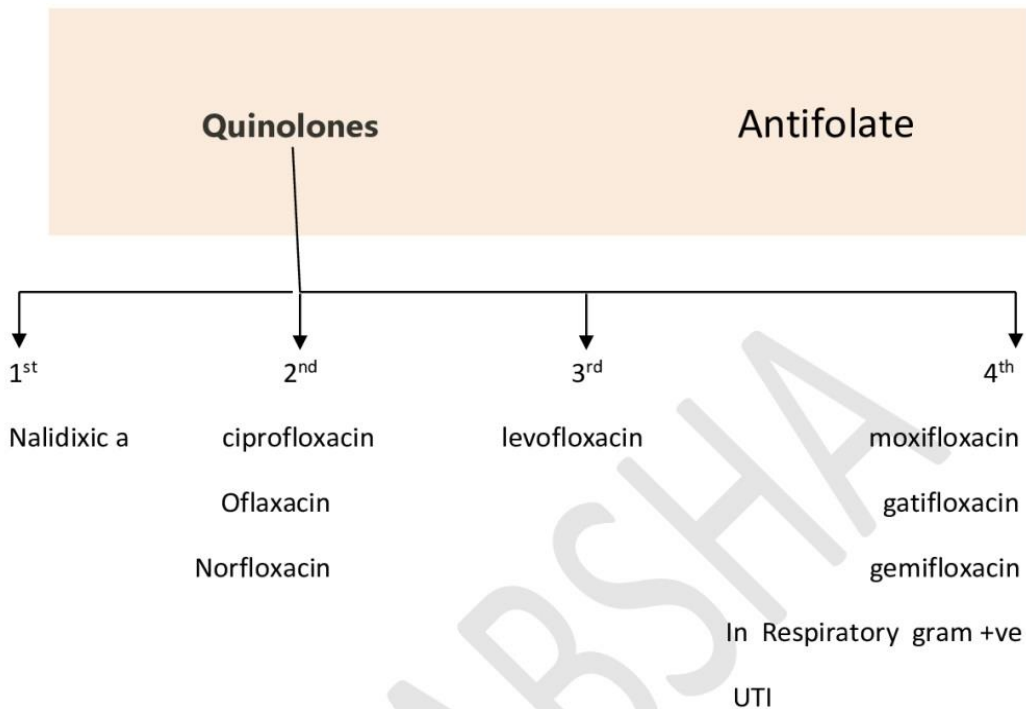
S/E diarrhea (pseudomembranous colitis)

Others

Mupirocin skin infection # MRSA

Protein synthesis inhibitor useful in impetigo

- Nucleic acid inhibitor



Topoisomerase II (gyrase)  # gram -ve (pseudomonas)

Topoisomerase IV  # gram +ve

Ciprofloxacin,, Ofloxacin inflamed meningitis

Ciprofloxacin and levofloxacin have enhanced activity against *Pseudomonas*.

Delafloxacin is active against **MRSA** and is the preferred quinolone if treating skin infections suspected to be caused by MRSA

All need renal adjustment except moxifloxacin (excreted by liver)

Moxifloxacin safe in renal but not liver

S/E QT prolongation

Photosensitivity like?

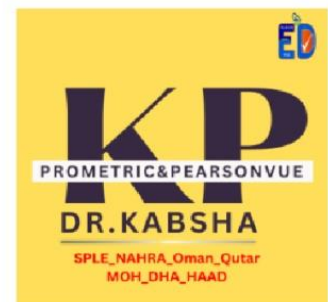
Articular cartilage erosion (tendon rapture)

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E, coli with CKD?

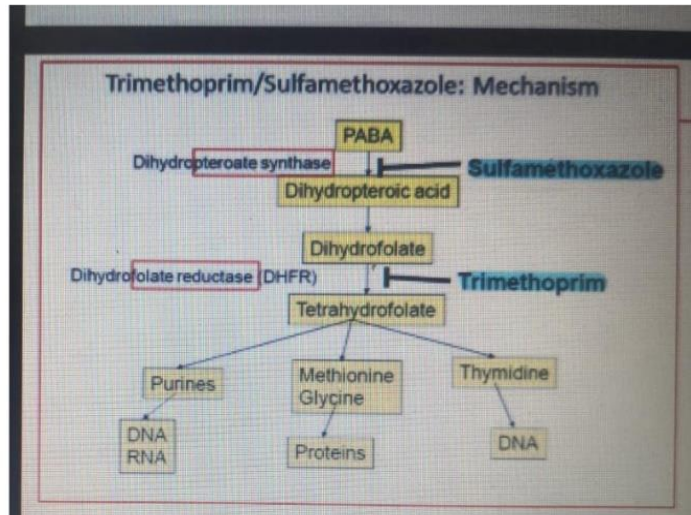
- a) Ciprofloxacin 400mg iv every 24 hrs
- b) Ciprofloxacin 400mg iv every 12 hrs

Drug	Age	cause
TTC, tigycyclin	<8 year	Tooth discoloration
Daptomycin	<12 year	Muscloskeletal problem
Quinolone	<18 year	QT prolongation tendon rapture



- **Sulfonamides** does not have activity against *Pseudomonas*, *Enterococci*, atypicals or anaerobes.

Sulfamethoxazole/ Trimethoprim synergistic



Sulfadiazine /pyrimethamine → toxoplasmosis

Sulfadoxine / pyrimethamine → malaria

Ttt of malaria ??

S/E

- Hyperkalemia ↑ k
- Sulfa allergy
- Hemolytic anemia (favism)
- Renal : crystalluria
- Newborn : kernicterus (bilirubin induced brain dysfunction)

Nitrofurantion

Used for uncomplicated UTI (cystitis only).

E. coli, Klebsiella, Enterobacter,

brown urine discoloration

pregnancy category B

contraindication

- G6PD deficiency (hymolytics anemia)
- Pregnancy (38-42 weeks)
- Neonate <1 month

Urinalysis can indicate if an infection is present, and is considered positive when ?

- there is evidence of pyuria (positive leukocyte esterase or > 10 WBC/mm³)
- bacteriuria ($\geq 10^5$ bacteria/mL in asymptomatic patients
- $\geq 10^3$ bacteria/mL in symptomatic males
- $\geq 10^2$ bacteria/ mL in symptomatic females and catheterized patients).
- asymptomatic bacteriuria in pregnancy).

The presence of bacteria alone is not indicative of a UTI and does not require treatment (exception: asymptomatic bacteriuria in pregnancy).

Bacteriuria ($\geq 10^5$ bacteria /mL on a urinalysis) in pregnant women must be treated even if asymptomatic. If not treated, bacteriuria can lead to pyelonephritis, premature birth and neonatal meningitis.

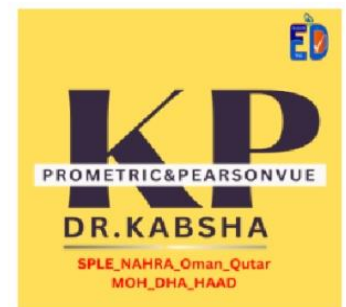
■ Beta-lactams are preferred (amoxicillin + clavulanate or an oral cephalosporin).

SMX/TMP can cause hyperbilirubinemia and kernicterus in the newborn if used close to delivery, and nitrofurantoin should be avoided in the 3rd trimester due to the possibility of hemolytic anemia in the infant}

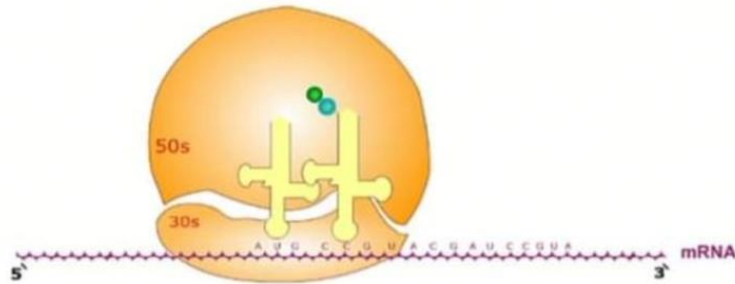
Notes

- Pregnant with UTI: **Nitrofurantoin**
- Pregnant with UTI + G6PD: **cefuroxime**
- Prevention recurrence of UTI: **TMP/SMX**
- Prophylaxis UTI: **Nitrofurantoin**
- Pregnant with E.coli and vaginal itching: **Nitrofurantoin**
- Pregnant with pyelonephritis she is been treated with IV ceftriaxone what is the most appropriate antibiotics: **Amoxicillin/ Clavulanate**
- Pregnant with syphilis: **Benzathine penicillin, Amox/Clavu**
- Breastfeeding with mastitis%: **dicloxacillin, Cephalexin, Amox / Clavu. # for 10-14 days # PCN allergy: Clarithromycin**

Prevent Respiratory synoptical virus (RSV): Palivizumab



Summary :



30s inhibition Prevent initiation

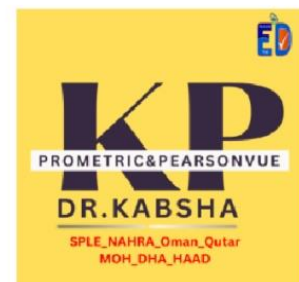
50s inhibition.... Prevent elongation

AMG, TTC30s

Macrolide, clindamycin50s

- **Do not require renal adjustment. Safe in renal :**

- Ceftriaxone
- Moxifloxacin
- Doxycycline
- Linezolid
- Clindamycin



Against pseudomonas:
Penicillin allergy:
In pregnancy:
Anti-staph from penicillin
Against MRSA:
Against VRSA:
Lipophilic:

Lipophilic ex:

- Quinolones
- Macrolides
- Rifampin
- Tetracyclines

- 1)  **vd** Large volume of distribution → excellent **tissue penetration**- including bone, lung and brain tissues
- 2) Hepatic metabolism → Potential for **hepatotoxicity** and drug – drug interaction
- 3) **Achieve intracellular concentration** → Active against atypical (intracellular) pathogens
- 4) **Excellent bioavailability**→IV.PO ratio is often 1:1

CAP:

OUT pt

Doxycylin

Macrolide

hospital

Doxycylin(Macrolide) + amoxicillin+cortisone

ICU

ampicilin +Q(macrolide)



Comorbid

Quinolone

Doxycylin(Macrolide) + amoxicillin



CAP(Community-Acquired Pneumonia)	
1- Empiric treatment of nonhospitalized patients	
a- Previously healthy and no antibiotic therapy in the past 3 months	A- Macrolide (clarithromycin or azithromycin if Haemophilus influenza is suspected) B- Doxycycline C- If local high-level macrolide resistance (minimum inhibitory concentration [MIC] of 16 mcg/mL or greater) exceeds 25%, use regimen from below.
b- Comorbidities (chronic obstructive pulmonary disease [COPD], diabetes mellitus, chronic renal or liver failure, congestive heart failure, malignancy, asplenia, or immunosuppression) or recent antibiotic therapy (within the past 3 months)	(a)Respiratory fluoroquinolone (moxifloxacin, gemifloxacin, or levofloxacin [750mg]) (b)Macrolide (or doxycycline) with high-dose amoxicillin (1 g three times/day) or amoxicillin/clavulanate (2 g twice daily) or a cephalosporin (ceftriaxone, cefuroxime, or cefpodoxime)
2- Empiric treatment of hospitalized patients with moderately severe pneumonia	
	i.Respiratory fluoroquinolone (moxifloxacin, gemifloxacin [oral only], or levofloxacin [750 mg]) ii.Ampicillin, ceftriaxone, or cefotaxime (ertapenem in select patients) plus a macrolide (or doxycycline) iii.Corticosteroids (unknown optimal agent, dose, or duration) decrease risk of mechanical ventilation and acute respiratory distress syndrome and decrease hospital stay by about 1 day ; greater incidence of hyperglycemia requiring treatment. Typical doses are prednisone or methylprednisolone 20–50 mg daily for 7 days.
3- Empiric treatment of hospitalized patients with severe pneumonia necessitating intensive care unit treatment (may need to add other antibiotics if Pseudomonas aeruginosa or methicillin-resistant S. aureus [MRSA] is suspected)	
	i. Ampicillin/sulbactam plus either a respiratory fluoroquinolone or azithromycin ii. Ceftriaxone plus either a respiratory fluoroquinolone or azithromycin iii. Cefotaxime plus either a respiratory fluoroquinolone or azithromycin

VAP(Ventilator-associated pneumonia)	
1- Empiric regimen should include antibiotics with activity against <i>S. aureus</i> , <i>P. aeruginosa</i> , and other gram-negative organisms	Piperacillin/tazobactam, cefepime, levofloxacin, imipenem or meropenem
2- Two antibiotics with activity against <i>P. aeruginosa</i> should be included if the patient has risk factors for MDR organisms (see below) or if <i>P. aeruginosa</i> resistance in the hospital unit to the antibiotic considered for monotherapy is greater than 10%	i. Options for first agent: Antipseudomonal β-lactam (ceftazidime, cefepime, imipenem, meropenem, piperacillin/tazobactam, or aztreonam) ii. Options for second agent: Aminoglycoside, fluoroquinolone (ciprofloxacin, levofloxacin) or colistin (aztreonam may be used with another β-lactam if no other second option is available)
3- An antibiotic with activity against MRSA should be included if the patient has risk factors for MDR organisms (see below) or if MRSA incidence in the hospital unit is greater than 10%–20%. Note: In locations where the prevalence of MRSA respiratory infections is low, a negative MRSA nasal screen suggests that MRSA antibiotics are unnecessary empirically. Locations with a high prevalence of MRSA respiratory infections should follow the rule above	i. Options if MRSA activity necessary: Vancomycin or linezolid ii. If an MRSA agent is used, then ceftazidime and aztreonam are potential alternative choices for the anti-pseudomonal agent
✓ Ideally, dose antibiotics based on pharmacokinetic/pharmacodynamic data, including antibiotic serum concentrations, extended infusions of β-lactams and weight based dosing of aminoglycosides ✓ Antibiotics should be de-escalated on the basis of culture results. Antibiotic therapy for <i>P. aeruginosa</i> can be de-escalated to one active agent on the basis of culture and sensitivity results. ✓ Treatment duration: 7-day course is recommended for all infections. ✓ Risk factors for MDR organisms: i. Intravenous antibiotic therapy within the past 90 days ii. Hospitalization of 5 days or more (VAP only) iii. Septic shock at time of VAP (VAP only)	

QUESTIONS:

1- A 39-year-old woman presents with complain of epigastria burning and pain for the last 6 month 33 / 272 significant medical history but reports an allergy to beta-lactam antibiotics. An office based Urea Breath Test is performed to diagnose Helicobacter pylori infection (see lab result).

Test	Result
Urea breath test	Positive

Which of the following is the most appropriate treatment regimen?

- A) Levofloxacin and lansoprazole for 14 days
- B) Omeprazole, clarithromycin, and amoxicillin for 10 days
- C) Omeprazole, clarithromycin, and metronidazole for 14 days**
- D) Omeprazole, bismuth subsalicylate, tetracycline, and azithromycin for 14 Days

2-A 55-year-old woman presents with symptoms of helicobacter pylori infections. She was previously treated with the classical amoxicillin, clarithromycin, omeprazole regimen. The physician is unable to obtain susceptibility testing for the Helicobacter pylori but the pattern of local antimicrobial resistance to penicillin's is well known. What would be the optimal treatment for her?

- A) Amoxicillin (1g), clarithromycin(500mg) and metronidazole(500 mg) plus omeprazole (20mg) BID for 14 days
- B) Amoxicillin (1g), clarithromycin (500 mg) and metronidazole (500 mg) plus omeprazole (40mg) BID for 14 days
- C) Bismuth subsalicylate 525 QID and tetracycline (500 mg) QID plus metronidazole (500 mg) and omeprazole (40mg) BID for 14 days**
- D) Bismuth subsalicylate (2) and doxycycline (100 mg) QID plus metronidazole (500 mg) TID and omeprazole (40 mg) BID for 14 days

3- Which of the following is the drug of choice for chlamydia infection?

- A. Amikacin
- B. Amoxicillin
- C. Tetracycline
- D. Azithromycin

4- Which of the following infections is prevented with the administration of 0.5% erythromycin eye ointment in newly born infants?

- A. mycoplasma conjunctivitis
- B. adenoviral conjunctivitis
- C. gonococcal ophthalmia
- D. chlamydial Infection

5-A 28-year-old pregnant female in labor is admitted to the hospital. She tested positive for Group B streptococcus colonization at her 36 week appointment. She has a penicillin allergy and her culture are resistant to both clindamycin and erythromycin. What is the recommended treatment for this patient?

- A) Cefazolin
- B) Penicillin G
- C) Ceftriaxone
- D) Vancomycin

6- Pt diagnosed with cellulitis take cephaloxin and no improvement..Next step take ?

- A) Vancomycin
- B) Gentamycin
- C) Ampicillin

7-Which is the best treatment for staphylococcus aureus positive mastitis

- A) Cephalexin
- B) Clindamycin
- C) Ciprofloxacin
- D) Nitrofurantoin

8-what is the recommended prophylactic therapy for pregnant women colonized with group B streptococcus in labor with penicillin allergies at high risk for anaphylaxis?

- A. Cefazoline
- B. Linezolid
- C. Penicillin G
- D. Clindamycin

9- Which of the following has similar antibacterial mechanism of penicillin?

- A. neomycin
- B. cephalexin
- C. tetracycline
- D. clindamycin

10- Which of the following is an indication for ciprofloxacin?

- A. Fever
- B. Arthropathy
- C. Tuberculosis
- D. Inhalational anthrax



11- Which of the following is the most appropriate empiric treatment for a patient with community-acquired pneumonia requiring admission to the hospital?

- A) Doxycycline and azithromycin
- b) Ceftriaxone and azithromycin
- c) Moxifloxacin and ceftriaxone
- d) Vancomycin and ceftriaxone

12- A 23-month-year-old, 14-kg infant, her mom brought her the hospital with signs and symptoms of bacterial meningitis. The baby attends day care and has a history of acute otitis media. A blood and CSF culture were ordered and pending. Which of the following is the empirical therapy for acute bacterial meningitis?

- A) Doxycycline plus linezolid
- B) Ampicillin plus gentamicin
- C) Cefuroxime plus ciprofloxacin
- D) Ceftriaxone or cefotaxime plus vancomycin

13- A 32-year-old male presents to an outpatient clinic with a 5-day history of productive cough. Purulent sputum and shortness of breath. He is diagnosed with community- acquired pneumonia (CAP). It is noted that this patient has a severe ampicillin allergy (anaphylaxis). Which of the following would be an acceptable treatment for this patient?

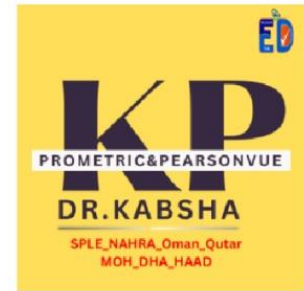
- A. Levofloxacin.
- B. Ciprofloxacin,
- C. Penicillin VK,
- D. Nitrofurantoin.

14- A 22-year-old female presents with a 2-day history of dysuria with increased urinary frequency and urgency. A urine culture and urinalysis are done. She is diagnosed with a urinary tract infection (UTI) caused by E. All of the following would be considered appropriate therapy for this patient except:

- A) Levofloxacin.
- B) Cotrimoxazole.
- C) Moxifloxacin.
- D) Nitrofurantoin

15-Someone got meningitis and his parents take him to hospital, how to protect others from getting infection?

- A) Siblings should take rifampicin
- B) Parents should take rifampicin
- C) Close contact with him should take rifampicin ☒
- D) Antibiotics not use for prophylaxis



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DR: KABSHA ...PROMETRIC&PEARSON VUECOURSE
00201064018621

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